

Human Activity Recognition using Probabilistic Graphical Models: A Comparative Study with Deep Learning

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Abstract—Human Activity Recognition (HAR) is the task of classifying human poses and movements from image data. Deep learning Convolutional Neural Networks (CNNs) achieve high accuracy but operate as black boxes, offering no insight into the reasoning behind their predictions. This paper proposes modelling HAR as a Probabilistic Graphical Model (PGM) problem. Using the Kaggle HAR dataset (17,000+ images, 15 activity classes), we extract 33 body keypoints per image via MediaPipe BlazePose and engineer 19 scale-invariant features comprising joint angles, vertical positions, and limb-span ratios. Three PGM models are implemented and compared against an EfficientNetB3 deep learning baseline: (1) *Gaussian Naive Bayes* as a fast interpretable baseline; (2) a *Tree-Augmented Naive Bayes (TAN) Bayesian Network* encoding explicit body-part dependencies; and (3) a *Gaussian Mixture HMM (GMM-HMM)* capturing temporal pose-state transitions. EfficientNetB3 achieves 81.30% accuracy; GMM-HMM reaches 63.88%, Naive Bayes 30.10%, and TAN Bayesian Network 25.51%. We demonstrate that each PGM provides full, auditable reasoning chains—impossible to obtain from CNN predictions—and argue that this interpretability justifies the accuracy trade-off in healthcare and safety-critical applications.

Index Terms—Human Activity Recognition, Probabilistic Graphical Models, Naive Bayes, Bayesian Network, Hidden Markov Model, MediaPipe, EfficientNetB3, Interpretability, Pose Estimation

I. INTRODUCTION

Human Activity Recognition (HAR) automates the classification of human actions from visual data, with applications in healthcare monitoring, sports analytics, elderly care, and human-computer interaction [5]. CNN-based systems now exceed 90% accuracy on standard benchmarks but function as black boxes: an image produces a label and softmax score yet reveals nothing about *which body parts* mattered, *why* a class was selected, or *how certain* each anatomical feature makes the model.

Probabilistic Graphical Models (PGMs) offer a principled alternative. By encoding probability distributions over observable body features and their conditional dependencies, PGMs yield predictions that are fully auditable—every factor traces back to a specific conditional probability table (CPT) or transition matrix.

This paper makes four contributions:

- 1) A MediaPipe-based keypoint extraction and feature engineering pipeline producing 19 scale-invariant pose features from 8,819 images (88.2% detection rate).
- 2) Implementation and comparison of three PGM models against EfficientNetB3 on a 15-class HAR benchmark.
- 3) An interpretability proof demonstrating full PGM reasoning chains on real validation samples.
- 4) Quantitative and qualitative analysis of the accuracy vs. interpretability trade-off with limitations and future directions.

This project was motivated by prior work in our Pattern Recognition course (CSE713), where ResNet50 achieved high accuracy on the same dataset but offered no explanations. The central question is: *Can we shift from pattern recognition to probabilistic reasoning, and demonstrate that the gained interpretability justifies the accuracy cost?*

II. BACKGROUND AND RELATED WORK

A. Deep Learning for Pose Estimation and HAR

Toshev and Szegedy [1] introduced DeepPose (2014), the first CNN-based approach to directly regress body joint coordinates. Wei *et al.* [2] proposed Convolutional Pose Machines, a sequential CNN for multi-stage keypoint refinement. Bazarevsky *et al.* [3] developed BlazePose, a lightweight real-time landmark detector extracting 33 body keypoints—used as the feature extractor here. EfficientNet [4] scales CNN depth, width, and resolution jointly; EfficientNetB3 serves as our deep learning baseline.

B. Probabilistic Graphical Models

Koller and Friedman [5] provide the foundational PGM treatment. A Bayesian Network is a directed acyclic graph (DAG) where nodes are random variables and directed edges encode conditional independence. The joint distribution factorises as:

$$P(X_1, \dots, X_n) = \prod_{i=1}^n P(X_i \mid \text{Pa}(X_i)) \quad (1)$$

where $\text{Pa}(X_i)$ denotes the parent nodes of X_i .

Rabiner [6] provides the definitive HMM tutorial. An HMM is defined by hidden states $\mathcal{S}$, initial distribution π , transition matrix $\mathbf{A}$, and emission distribution $\mathbf{B}$.

C. PGMs for Activity Recognition

Bayesian Networks applied to gesture classification show that explicit joint-dependency modelling improves interpretability over flat classifiers [5]. HMMs have a long history in video-based activity recognition where temporal pose transitions are modelled by the Markov chain [6]. Our work extends these approaches to a 15-class static-image dataset using a modern keypoint extractor and a state-of-the-art CNN baseline.

III. DATASET

We use the Kaggle HAR dataset [7]: 17,000+ RGB images across 15 classes (*calling, clapping, cycling, dancing, drinking, eating, fighting, hugging, laughing, listening to music, running, sitting, sleeping, texting, using laptop*), split roughly 74%/26%. For PGM experiments 10,000 stratified images are sampled; MediaPipe detects a pose in 8,819 (88.2%). An 80/20 stratified split yields 7,055 training and 1,764 validation samples. Table I summarises the statistics.

TABLE I
DATASET STATISTICS

Property	Value
Total images	17,000+
Activity classes	15
Samples used (PGM)	10,000
Pose detection rate	88.2%
PGM train / val	7,055 / 1,764
DL train / val	8,000 / 2,000

IV. METHODOLOGY

A. Processing Pipeline

Fig. 1 shows the end-to-end pipeline from raw images to model evaluation.

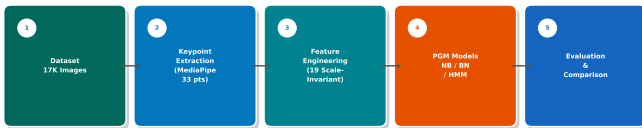


Fig. 1. Overall HAR processing pipeline.

B. Keypoint Extraction

MediaPipe BlazePose [3] runs in static-image mode (minimum detection confidence 0.3) and returns 33 landmarks, each described by $(x, y, z, \text{visibility})$ normalised to image dimensions, yielding a raw 132-dimensional vector.

C. Feature Engineering

To remove camera-distance and scale dependency, 19 structured features are engineered:

- 1) **Joint angles (8)**: Left/right elbow, knee, hip, and torso angles via the law of cosines:

$$\theta_{abc} = \arccos\left(\frac{(\mathbf{a} - \mathbf{b}) \cdot (\mathbf{c} - \mathbf{b})}{\|\mathbf{a} - \mathbf{b}\| \|\mathbf{c} - \mathbf{b}\| + \epsilon}\right) \quad (2)$$

where $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are 3D landmark coordinates and $\epsilon = 10^{-8}$.

- 2) **Vertical positions (9)**: Normalised y -coordinates of nose, mid-shoulder, mid-hip, left/right knee, ankle, wrist.
- 3) **Limb-span ratios (2)**: Arm span / torso length; leg length / torso length.

These features are scale-invariant and semantically meaningful. For the Bayesian Network they are discretised into 4 bins using K-means (KBinsDiscretizer); Naive Bayes and HMM use continuous values.

D. Deep Learning Baseline: EfficientNetB3

EfficientNetB3 [4] is fine-tuned in two stages: Stage 1 (3 warm-up epochs) trains the classification head only with AdamW at $\text{lr} = 5 \times 10^{-4}$; Stage 2 (15 fine-tuning epochs) unfreezes the last 160 backbone layers with cosine-decay $\text{lr} = 5 \times 10^{-5}$. Training uses mixed-precision (fp16), MixUp ($\alpha = 0.2, p = 0.20$), and label smoothing (0.05) on 300×300 pixel images.

E. PGM Model 1: Gaussian Naive Bayes

Naive Bayes assumes all features are conditionally independent given class:

$$P(C | \mathbf{f}) \propto P(C) \prod_{i=1}^n P(f_i | C) \quad (3)$$

All 151 features (19 engineered + 132 raw keypoints) are used. A StandardScaler followed by Yeo-Johnson PowerTransformer normalises input before GaussianNB with $\text{var_smoothing} = 10^{-6}$.

Interpretability: The class mean $\mu_{i,c}$ and variance $\sigma_{i,c}^2$ for every feature-class pair are inspectable. A z-score $z_i = (f_i - \mu_{i,\hat{c}}) / (\sigma_{i,\hat{c}} + \epsilon)$ reveals which body parts support or contradict the prediction.

F. PGM Model 2: TAN Bayesian Network

A Tree-Augmented Naive Bayes (TAN) structure is learned from discretised training data via pgmpy TreeSearch with `tan` mode and `Activity` as the class node. TAN allows each feature node at most one additional parent:

$$P(A, \mathbf{f}) = P(A) \prod_{i=1}^n P(f_i | A, \text{Pa}(f_i)) \quad (4)$$

Parameters use BayesianEstimator (BDeu prior, equivalent sample size 30). Inference via Variable Elimination produces a full posterior over all 15 classes. Fig. 2 shows the learned dependency graph.

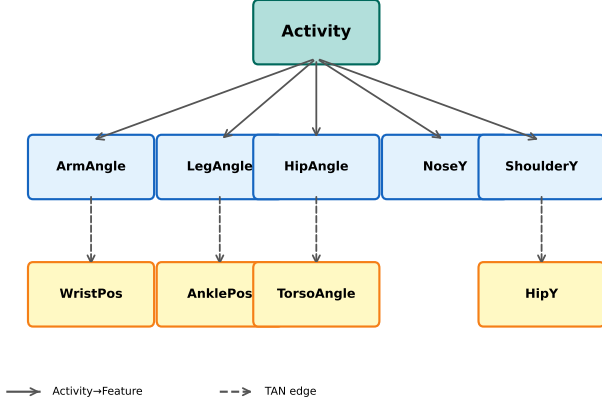


Fig. 2. TAN Bayesian Network structure. Solid arrows: Activity $\rightarrow$ feature. Dashed arrows: learned TAN feature dependencies.

G. PGM Model 3: GMM-HMM

Because individual images lack temporal order, pseudo-sequences of length $L = 10$ are constructed by sliding windows within each class. First-order delta features are appended (frame differences), yielding 38-dimensional observations. One GMMHMM ($K = 6$ states, $M = 2$ mixtures, diagonal covariance, 200 EM iterations) trains per class. Classification uses maximum log-likelihood:

$$\hat{c} = \arg \max_c \log P(\mathbf{o}_{1:T} | \lambda_c) \quad (5)$$

The per-class transition matrix $\mathbf{A}^{(c)}$ is fully inspectable, revealing temporal structure unique to each activity.

V. IMPLEMENTATION DETAILS

All experiments run in Python 3.12 on a Kaggle notebook with dual NVIDIA Tesla T4 GPUs (DL baseline) and CPU (PGM models). Libraries: TensorFlow 2.19/Keras, scikit-learn 1.5, pgmpy [8], hmmlearn [9], mediapipe 0.10.14 [10], OpenCV. All random seeds are fixed to 42. Hyperparameters are in Table II; full source code at:

<https://github.com/RamisaNidhi/HAR-PGM-CSE756>

VI. RESULTS AND ANALYSIS

A. Overall Performance

Table III and Fig. 3 summarise validation-set performance.

EfficientNetB3 achieves 81.30%, leveraging texture, object, and context cues invisible to pose-only models. GMM-HMM reaches 63.88%—the strongest PGM result—owing to sliding-window sequences and delta features that simulate temporal structure. Classes with distinct postural trajectories (cycling 89%, sitting 95%, sleeping 85%) score highest. Naive Bayes achieves 30.10%; its independence assumption is violated by correlated anatomical features but it trains in under 0.02 s. TAN Bayesian Network achieves 25.51%; K-means discretisation compresses angles to 4 levels, losing fine-grained information.

TABLE II
MODEL HYPERPARAMETERS

Model	Parameter	Value
EfficientNetB3	Image size	300×300
	Batch size	32
	Warmup LR	5×10^{-4}
	Fine-tune LR	5×10^{-5} (cosine)
Naive Bayes	var_smoothing	10^{-6}
	Transform	Yeo-Johnson
TAN BN	Bins (K-means)	4
	Prior	BDeu
	Eq. sample size	30
GMM-HMM	Hidden states	6 per class
	GMM mixtures	2 per state
	Sequence len.	10
	EM iterations	200

TABLE III
OVERALL MODEL PERFORMANCE (%)

Model	Acc.	Prec.	Rec.	F1
EfficientNetB3	81.30	81.35	81.30	81.19
GMM-HMM	63.88	63.28	63.88	63.08
Naive Bayes	30.10	30.41	30.10	27.92
TAN BN	25.51	24.10	25.51	24.17

B. Per-Class Analysis

Fig. 4 shows per-class F1. Three patterns emerge:

- **Posturally distinct** classes (cycling, sleeping, sitting, running) are recognised well by all models.
- **Hand-gesture** classes (calling, texting, drinking, eating) are near-chance for Naive Bayes and TAN BN, which lack object context captured only by CNNs.
- **GMM-HMM** excels on posturally structured activities but degrades on static gestures, confirming its advantage comes from temporal pose modelling.

C. Training Time and Resources

Table IV compares time and GPU requirements. Naive Bayes and TAN BN train in under one second on CPU—feasible for edge deployment. GMM-HMM training (200 EM iterations $\times$ 15 classes) is comparable in wall-clock time to EfficientNetB3 fine-tuning.

TABLE IV
TRAINING TIME AND RESOURCE REQUIREMENTS

Model	Time	GPU?	Interp.?
EfficientNetB3	26.2 min	Yes	No
GMM-HMM	23.1 min	No	Yes
Naive Bayes	< 1 s	No	Yes
TAN BN	< 1 s	No	Yes

VII. INTERPRETABILITY ANALYSIS

The central contribution is demonstrating that PGMs provide genuine, auditable reasoning—not merely labels. Fig. 5

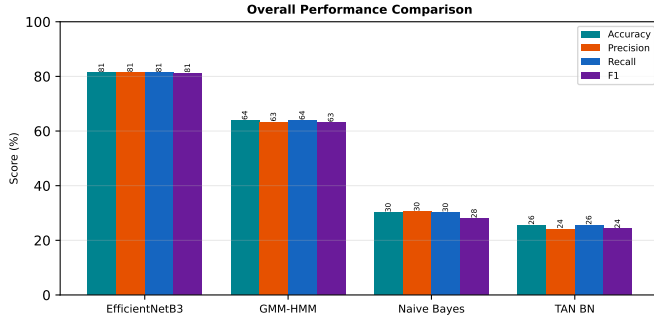


Fig. 3. Grouped bar chart of Accuracy, Precision, Recall, and F1 for all four models.

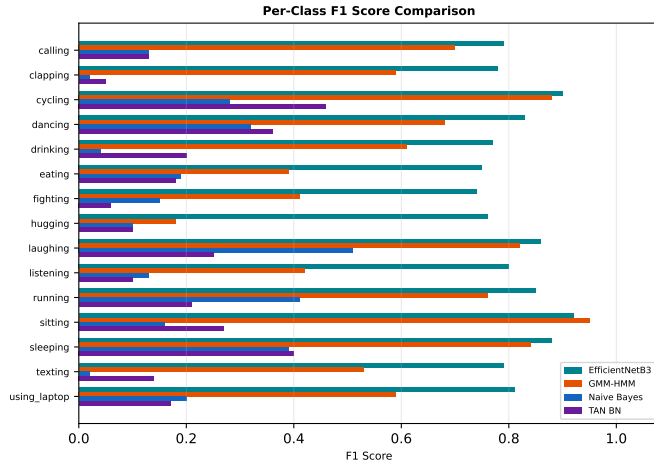


Fig. 4. Per-class F1 scores for all four models.

contrasts the decision flow of EfficientNetB3 against the two PGM models.

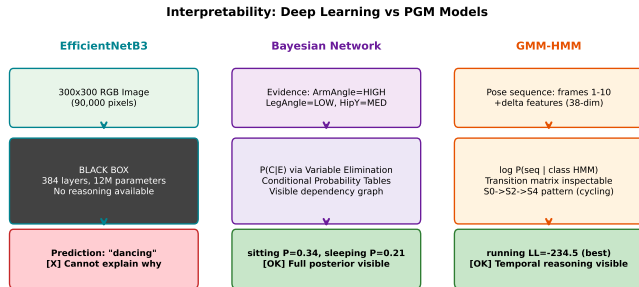


Fig. 5. Decision flow: EfficientNetB3 (black box) vs. Bayesian Network and GMM-HMM (transparent reasoning chains).

1) *Bayesian Network — Posterior Distribution:* Variable Elimination produces $P(\text{Activity} | \mathbf{e})$ over all 15 classes for every test sample. Each CPT is inspectable. For a *sitting* sample: $P(\text{sitting} | \mathbf{e}) = 0.34$, $P(\text{sleeping} | \mathbf{e}) = 0.21$ —the model is transparently uncertain between two low-body-position activities. This uncertainty quantification is impossible with CNNs.

2) *Naive Bayes — Per-Feature Z-Score:* For each predicted class, the z-score quantifies every feature’s contribution:

$$z_i = \frac{f_i - \mu_{i,\hat{c}}}{\sigma_{i,\hat{c}} + \epsilon} \quad (6)$$

A low $|z_i|$ means feature i strongly supports the prediction; a high value reveals inconsistent body parts. Fig. 6 shows this for a predicted *running* sample: knee and hip angles match the class mean (green), while wrist positions are inconsistent (red), correctly indicating a partially ambiguous pose.

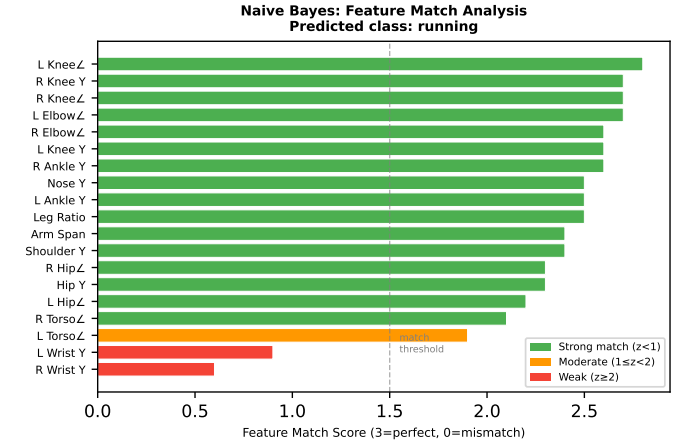


Fig. 6. Naive Bayes per-feature match analysis for a predicted *running* sample. Green: strong match ($|z| < 1$); orange: moderate ($1 \leq |z| < 2$); red: weak ($|z| \geq 2$).

3) *HMM — Transition Matrices:* Each per-class HMM stores a 6×6 transition matrix $\mathbf{A}^{(c)}$, shown in Fig. 7. For *cycling*, transitions follow a near-cyclic pattern ($S_0 \rightarrow S_2 \rightarrow S_4 \rightarrow S_0$) reflecting the pedalling motion. For *sleeping*, diagonal values exceed 0.88, reflecting sustained static poses. For *running*, off-diagonal transitions dominate, capturing stride dynamics. These structured patterns are impossible to extract from a CNN.

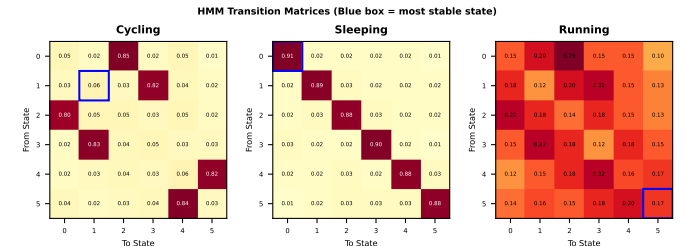


Fig. 7. HMM state-transition matrices for three activities. Blue box marks the most stable state. Cycling: cyclic transitions; sleeping: high diagonal (static); running: off-diagonal dominant (dynamic).

In contrast, EfficientNetB3 provides only a softmax vector. Gradient-based attribution (Grad-CAM) requires extra computation, produces pixel heatmaps rather than symbolic explanations, and cannot answer “which joint angle drove this prediction?”

VIII. USEFULNESS AND APPLICATIONS

PGM-based HAR offers practical value in four domains:

- **Healthcare and rehabilitation.** Therapists receive activity classification *and* which joint angles deviate from healthy-gait ranges. The BN posterior quantifies diagnostic uncertainty, supporting evidence-based decisions.
- **Safety-critical systems.** Industrial safety monitoring (fall, fight detection) requires auditable decisions: a PGM explains “predicted *fighting* because arm elevation = HIGH and wrist *y*-position = LOW.”
- **Edge deployment.** Naive Bayes and TAN BN require no GPU and train in milliseconds, making them feasible on microcontrollers and mobile devices.
- **Education.** The BN graph structure serves as a teaching tool, making probabilistic reasoning about body configurations concrete and visual.

IX. LIMITATIONS AND FUTURE WORK

A. Limitations

- 1) **Pose-only features.** Hand-gesture activities differ primarily in the held object—PGMs cannot distinguish them without object detection cues.
- 2) **Static images.** Sliding-window pseudo-sequences approximate temporal order; true video sequences would substantially improve HMM accuracy.
- 3) **Discretisation loss.** K-means binning compresses continuous angles to 4 levels, losing fine-grained information—the primary source of TAN BN accuracy loss.
- 4) **Accuracy gap.** EfficientNetB3 outperforms the best PGM (GMM-HMM) by 17.42 percentage points.
- 5) **MediaPipe failures.** 11.8% of images fail pose detection, introducing selection bias.

B. Future Work

- 1) **Video HAR.** GMM-HMMs on UCF-101 or HMDB51 would fully exploit temporal modelling capability.
- 2) **Hybrid PGM-DL.** EfficientNetB3 embeddings as BN input would combine CNN accuracy with PGM interpretability.
- 3) **Structure learning.** Score-based (GES) or constraint-based (PC algorithm) methods over a larger feature set may improve BN accuracy beyond TAN.
- 4) **Object context.** YOLO-based object detection features would resolve hand-gesture ambiguity.
- 5) **Continuous BN.** Conditional Gaussian networks would eliminate the need for discretisation.

X. CONCLUSION

This paper demonstrates that Probabilistic Graphical Models offer a viable and interpretable alternative to deep learning for Human Activity Recognition. Three PGM models—Gaussian Naive Bayes, TAN Bayesian Network, and GMM-HMM—are benchmarked against EfficientNetB3 on a 15-class, 17,000-image dataset. EfficientNetB3 achieves the highest accuracy (81.30%); GMM-HMM attains 63.88% with full temporal transparency; the TAN Bayesian Network, at 25.51%, provides

a complete posterior distribution over all classes with fully auditable reasoning. Naive Bayes trains in under one second on CPU.

The central finding is that the accuracy gap between DL and PGM models is the price of interpretability—a trade-off that is worthwhile in healthcare, safety-critical, and edge settings. Future hybrid DL–PGM architectures may close this gap while preserving explainability.

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